Operator: Aera Energy LLC
Well: 377K-20
Type: Injector
Wellbore: 00
API no.: 0402976993

Wellbore Schematic

Datum: KELLY BUSHING @ 579.00ft (default)
Ground Elev: 569.00ft
PBMD: 1,320.00ft
TD MD: 1,547ft
Orientation: Vertical
Completion Date: MM/DD/YYYY

CASING

Size(in)	Top MD(ft)	Base MD(ft)
18.000	10	40
10.750	10	1,405
8.625	10	788
7.000	1,343	1,547
6.625	740	1,319

CEMENT STAGES

Job Type: Primary Date: 11/24/1985

Csg Size	Hole Size(in)	Top MD(ft)	Btm MD(ft)
10.750	14.750	10	1,405

Job Type: Plug Date: 1/12/2009

Csg Size	Hole Size(in)	Top MD(ft)	Btm MD(ft)
7.000	7.000	1,350	1,547

REMARKS

Cemented 10.75" casing lead slurry with 576 sacks of Class "G" cement, 35% Silica Flour, 2% gel, 3% calcium chloride, 1/2# per sack of flocele.
Tailed in with 118 sacks of Class "G", 35% Silica Flour, 3% calcium chloride, 1/2# per sack of flocele.15.5 bbls of cement returns to the surface.

Cemented Plugback from 1547' to 1350' with 25.4 sacks of 15.7 ppg Class "G" Cement with Yield 1.53 cuft/sk, mix water ratio 6.35 gal/sk, 35% Silica Flour, no returns to surface.

MARKER DETAILS

Name	MD top(ft)
Tulare	404.13

PLUG DETAILS

Assembly Name: Cement Plug

Type	Top MD(ft)	Btm MD(ft)
Cement Plug	1,350	1,547

Casing/Perfs/Slots Details	Schematic	Hole Details
Ground Level		
CSG:18.000in, , , , 10.00ft, 40.00ft		
CSG SHOE @ 40ft		24.000in to 50.00ft
CSG:8.625in, 32.00ppf, , , 11.00ft, 787.11ft		
CSG:6.625in, 24.00ppf, , , 743.58ft, 787.08ft		
SLOTS:0.012, 1.250in, 6, 6.000, 787.08ft, 1,317.08ft		
CSG SHOE @ 788ft		
PERFORATED, PERFORATED, 808ft, 882ft, 75shot/ftSPF		
PERFORATED, PERFORATED, 892ft, 904ft, 12shot/ftSPF		
PERFORATED, PERFORATED, 926ft, 934ft, 8shot/ftSPF		
CSG:10.750in, 40.50ppf, K-55, , 10.00ft, 1,369.00ft		
PERFORATED, PERFORATED, 1,040ft, 1,052ft, 12shot/ftSPF		
PERFORATED, PERFORATED, 1,068ft, 1,092ft, 24shot/ftSPF		
PERFORATED, PERFORATED, 1,102ft, 1,122ft, 20shot/ftSPF		
PERFORATED, PERFORATED, 1,136ft, 1,158ft, 24shot/ftSPF		
PERFORATED, PERFORATED, 1,174ft, 1,226ft, 48shot/ftSPF		
PERFORATED, PERFORATED, 1,240ft, 1,252ft, 12shot/ftSPF		
CSG:6.625in, 24.00ppf, , , 1,317.08ft, 1,319.28ft		
CSG SHOE @ 1,319ft		
CSG:7.000in, 31.00ppf, K-55, , 1,347.00ft, 1,394.00ft		
Insert Valve @ 1,369ft		14.750in to 1410.00ft
CSG:10.750in, 40.50ppf, K-55, , 1,369.00ft, 1,405.00ft		
SLOTS:0.025, , 0, , 1,394.00ft, 1,535.00ft		
CSG SHOE @ 1,405ft		19.000in to 1547.00ft
CSG:7.000in, 31.00ppf, K-55, , 1,535.00ft, 1,547.00ft		9.625in to 1547.00ft
CSG SHOE @ 1,547ft		

Initials: xyz Date: mm/dd/yyyy

Well Notes: